

Kalshi LAX Offline Research Goals

Version 1.3 | Prepared September 24 2026 Pacific time | Implementation status updated during execution

Workspace: C:\Users\darks\Documents\Codex\kalshi_weather_llm

This document is the implementation contract for a historical, offline research system for Kalshi daily maximum temperature contracts at Los Angeles International Airport. It supersedes earlier suggestions to build an online recorder, prospective live feed, or general-purpose research framework. The latest user instruction controls: obtain historical data, analyze it offline, and keep the trading system offline until the user has reviewed the results and separately requests a change.

1 Mission and the two goals

Our research target is to identify opportunities with at least 10 percent estimated net expected return per trade on total entry outlay, then determine whether the selection policy produces credible positive out-of-sample historical returns. We will report the size of the estimated edge, historical simulated realized returns, uncertainty, opportunity count, and execution limitations separately. Completing this project does not require manufacturing a profitable result.

Goal 1 is to build and verify the complete offline research foundation. Obtain and audit the preferred collector; acquire the historical inputs and settlement outcomes; preserve their provenance; implement comparable baselines, calibrated probabilities, a historical market replay, deterministic evaluation, and the bounded research controller. Finish with a signed-by-software readiness report and reproducible commands. The completion label is `READY_FOR_OFFLINE_CAMPAIGN` only when the required checks pass.

Goal 2 is to launch and complete the first bounded offline multi-agent research campaign. Use the verified foundation to run independent investigations, exchange evidence, combine promising findings, reproduce results, challenge candidates, and evaluate a frozen candidate when justified. Save an inspectable historical performance report and campaign record. The completion label is `OFFLINE_CAMPAIGN_COMPLETE`; its scientific conclusion may be improvement, no improvement, or insufficient evidence.

Goal 2 depends on Goal 1. A queued task or a collection of prompts is not a completed campaign. Neither goal includes live prices, live or paper orders, unattended online collection, or deployment. Offline ledger entries are hypothetical backtest transactions only.

Current status: Goal 1 is in progress. Historical acquisition, source audits, normalization, baseline models, the development replay and a bounded controller are implemented and locally tested. A pinned local OpenAI gpt-oss-20b worker passed an actual tool-free capability probe. Full weather acquisition, verified historical baselines and Goal 1 readiness remain incomplete, so Goal 2 has not launched. `README.md` and `docs/IMPLEMENTATION_STATUS.md` contain the execution record. The registered first-campaign protocol is in `docs/CAMPAIGN_PROTOCOL.md`. No profitability has been demonstrated.

The original design source is the user's Apply Navier Stokes Method conversation [S1]. We retain independent exploration, local collaboration, synthesis, resource reallocation, and separate verification. The exact roles, budgets, and evaluation rules below are our design for Kalshi.

2 What offline means for this project

There are three separate execution lanes. The **acquisition lane** may use the internet to download historical weather forecasts, observations, archived market data, rules, and source documentation. These are finite, resumable jobs for specified historical dates. They must not subscribe to a live feed or run indefinitely.

The **experiment lane** reads only frozen local inputs. Forecast generation, feature construction, calibration, scoring, market replay, and report calculations must run with network access denied. A missing file produces a clear failure or an explicitly registered missing-data treatment; it must never trigger a hidden fetch of today's data.

The **agent lane** now uses a pinned local OpenAI gpt-oss-20b model through a standalone completion executable. Workers receive curated development packets and no file, command, network or application tools. An actual model probe and host-owned negative capability tests passed. The native runtime and host curator remain trusted; this is not an operating-system sandbox. The earlier hosted Codex CLI worker path remains disabled because its tested isolation evidence was insufficient. Ordinary work in this Codex conversation still uses connectivity, while the research model inference and historical experiments run locally.

Enforce these boundaries in code and available process permissions. Provide separate acquisition and experiment entry points. The experiment runner must not import or call download functions. Use a frozen source manifest and artifact hashes to prove which files were consumed. Test that an offline experiment succeeds with its inputs and fails cleanly when one is removed.

No scheduler, heartbeat, Windows scheduled task, WebSocket subscription, live prediction job, broker connection, or order endpoint should be enabled by either goal. Source credentials, if needed, are restricted to authorized historical read access and never printed or committed. Existing access can be reused; do not invent account credentials or purchase subscriptions.

Future online operation requires a new user instruction after the historical results are reviewed. A passing readiness test, positive backtest, or completion of Goal 2 does not grant that authorization.

3 Assets verified and access status

The preferred dependency is https://github.com/ciel-ravencourt/weather_data_collector.git. Its signed-in GitHub page was inspected at commit a82f8aee0afecf568b3ce16f339c0bd7201773b8. The repository is private. Its README describes GFS, GFS Seamless, NAM Grid 218, and NBM, daily JSON output, raw GRIB2 caching, and a fixed 2 degrees Fahrenheit Gaussian standard deviation [S2]. These are README-observed capabilities, not execution-verified claims.

Collector access was resolved during implementation through the signed-in browser's normal individual-file downloads. Nine source files are now preserved untouched under external/weather_data_collector/ with a source manifest and hashes. This is a source snapshot, not a Git clone, and excludes Git history and example data. The full audit is in docs/COLLECTOR_AUDIT.md. The native Windows adapter uses ecCodes; no operating system or browser security setting was changed.

The original collector's probability calculation and JSON schema have also been executed with networking denied on two explicitly synthetic temperature fixtures. Its weighted discrete mixture differs from our equal-blend single-Gaussian diagnostic. Original historical four-feed outputs have not been recovered; the saved synthetic originals verify calculation compatibility, not historical forecast quality. docs/COLLECTOR_REFERENCE.md records the comparison and its limits.

Three historical starter files were downloaded into data/raw/bootstrap/. The NOAA file ncei_klax_daily_2020_2025.csv contains 2,192 station-day rows from 2020-01-01 through 2025-12-31, with no blank TMAX field. The station identifier was checked as USW00023174. This is a schema and completeness check, not a complete quality or settlement reconciliation audit.

The Kalshi file kalshi_lax_historical_markets_sample.json contains the first 100 returned historical KXHIGHLAX contracts and has a remaining pagination cursor. It is not the full series history. One inspected contract identifies Los Angeles Airport and the NWS daily climatological report as the settlement basis. Its result and expiration value are outcome data and must not become forecast inputs.

kalshi_historical_cutoff.json records the API archival boundaries at retrieval time.

docs/bootstrap-download-manifest.json records source URLs, retrieval times, byte counts, hashes, and basic checks. Downloads occurred September 25 UTC, still September 24 Pacific time.

Implementation update: all 2,166 historical KXHIGHLAX contracts for January 5 through December 31, 2025 and 72,390 hourly price records are downloaded and hash-verified. Archived NWS CLILAX products are downloaded, versioned and compared with settlement records. Historical GFS/NBM fields for 2024-2025 are being acquired in a finite resumable job. Historical depth remains unavailable. The current baseline uses a named assumed-fill scenario and cannot establish executable gains. The earlier bootstrap sample is retained for provenance, but is no longer the full market input.

4 The architecture to implement

Build one Kalshi-specific Python project with clear boundaries. Prefer local Parquet files for normalized tables and SQLite for task state and registries. Use immutable files for raw inputs and large artifacts. Pin dependency versions and random seeds. Choose a supported GRIB environment after auditing the collector rather than assuming that the existing Windows Python can load its native dependencies.

The acquisition adapter wraps the supplied collector and supplementary source clients. The normalizer turns provider formats into canonical records without discarding issue times, valid times, source versions, or quality flags. The dataset builder creates frozen chronological partitions. The forecasting layer produces distributions; the contract mapper converts those distributions into YES and NO probabilities.

The historical replay engine applies a fixed decision policy to time-ordered market evidence. It handles prices, costs, quantities, opportunity selection, hypothetical entries, and settlement. The evaluator computes metrics from saved predictions and ledger entries. The report generator explains results and limitations.

The research controller assigns bounded tasks to agents, validates their proposed actions, manages dependencies and budgets, and stores evidence. Workers create candidate implementations in isolated work areas. They cannot alter protected scoring code, change the research target, increase budgets, or access the final test labels.

The authoritative workflow is:

```
Historical acquisition -> immutable raw files -> normalized tables
                                     -> frozen dataset partitions
Research task -> candidate code -> offline experiment runner
                -> predictions and replay ledger -> deterministic metrics
                -> evidence registry -> synthesis -> next research tasks
Frozen candidate -> separate final evaluator -> historical results report
```

Implement the minimum reliable version of each module before adding complexity. JSON schema validation, file hashes, transactional task state, and a small task queue are adequate foundations. A distributed platform, vector database, or custom general-purpose agent framework is not a prerequisite.

5 Integrating the preferred collector

Preserve an untouched source snapshot and record its commit, license or usage terms, requirements, and file hashes. Place it under `external/weather_data_collector/`; keep bulk external source snapshots outside the main repository's versioned content unless deliberately vendored with attribution. Record local patches separately and identify their parent revision.

The README's entry point is `scripts/generate_dataset.py`. It accepts a city, one date or a date range, a UTC initialization, and a strictness option. It documents Python 3.12 or later, `pygrib`, `ecCodes`, and a preference for WSL or Linux. Treat those instructions as starting points for an environment check; do not silently install an operating system or require a reboot.

Station selection must be corrected or explicitly validated. A generic Los Angeles city lookup may resolve a downtown location rather than the airport. The adapter must accept the verified station coordinates and retain chosen model grid cells and interpolation rules. Evaluate neighboring cells if coastal land-water representation matters, using development data only.

Time and interval semantics require a code audit. The collector describes local calendar days. The target is the contract-defined climate day. Its GFS methods use interval maximum fields; NAM uses sampled three-hourly temperatures; NBM uses sampled hourly temperatures. An interval maximum that crosses a target-day boundary cannot be apportioned exactly without additional information. Sparse samples can miss a daily peak. Document these differences and compare like-for-like targets.

Probabilities must be recalibrated. Preserve the fixed 2 degree Gaussian as an explicitly naive baseline. Learn bias and uncertainty from past forecast errors, separately from evaluation data. GFS and GFS Seamless must not be counted as independent evidence without examining how they overlap.

Add acquisition timestamps, as-of availability, content hashes, coverage checks, explicit missing-model statuses, and read-only cached execution. Never infer that an initialization timestamp proves the data was publicly available then. Verify the filename, URL, horizon, and units for a tiny historical batch before scaling.

6 Historical weather datasets

Acquire the following classes, with provenance and quality checks. Prefer authoritative providers and reuse the collector when its outputs meet the requirements.

- **Official target outcomes:** the relevant NWS daily climate reports and Kalshi's settled outcomes. Preserve raw report text, issue time, revisions, station, climate date, daily high, and source links. Historical NCEI daily summaries support climatology and reconciliation; they are not automatically interchangeable with the contract settlement label.
- **Intraday station observations:** timestamped KLAX temperature, dew point, wind, pressure, sky cover and ceiling where available. Prefer NOAA archival sources; a documented archive such as Iowa Environmental Mesonet may be used with explicit source and quality qualifications. Preserve original reports, precision, flags, and corrections.
- **Archived forecasts:** start with the collector's NBM, GFS and NAM paths. Test HRRR as an additional short-range candidate through a documented archive [S6, S7]. No model is designated the best before a same-date, same-cutoff comparison.
- **Forecast uncertainty:** use archived ensemble members or native probabilistic fields when available and affordable. Otherwise fit distributions from genuinely prior forecast errors. GEFS is an optional challenger, not a dependency for the first working baseline.
- **Context features:** historical NWS forecast discussions, cloud observations, coastal winds and pressure gradients. Admit a feature only when its historical release time can be established or conservatively bounded.

For every forecast record retain provider, model and version, initialization, forecast lead, valid time or interval, variable, units, grid geometry, extraction method, source object, and availability evidence. Keep forecast files separate from analysis or reanalysis products. Reanalysis may help diagnosis, but cannot substitute for forecasts that existed at the simulated decision time.

Prefer point or small-region extraction around the verified airport. Retain the source object's checksum and extraction recipe. If source files cannot be retained within storage limits, record the retention policy and how an auditor can obtain the same bytes; never imply exact reproducibility when the provider can revise them.

7 Historical Kalshi datasets

Discover the exact daily-high series and relevant event identities from metadata. KXHIGHLAX is the verified starter query, not permission to infer every contract's station or rules from its ticker. Preserve archived market and event metadata, threshold definitions, contract range inclusivity, open and close times, outcome, settlement value, and rules [S3, S4].

Download complete historical pagination for the selected period. Kalshi separates older and recent records using archival cutoffs; an endpoint named live may still be needed to retrieve a settled historical record. Requests must be bounded to past settled events. Never call a latest-price feed simply to fill a historical gap.

Acquire timestamped trades and available bid and ask candle fields, with their documented aggregation meaning. Preserve timestamps, side conventions, units, sizes, and any block-trade flags. Historical trades are records of someone else's executions; they do not prove our hypothetical order would have filled. A candle high or low is not an executable quote at every instant of that candle.

Seek historical order-book snapshots or depth only through an authorized source with documented coverage. The official historical endpoints inspected here establish markets, candles and trades; they do not establish that a full historical order book is available for this study. Do not promise it. Paid datasets require a separately agreed purchase and budget.

Collect fee schedules effective during the historical period, relevant market-specific adjustments, and evidence of rule changes. Today's fees may be used as a clearly labeled present-cost sensitivity case, not silently substituted for historical fees [S5]. If exact historical costs are unavailable, provide conservative sensitivity bounds and downgrade the economic conclusion.

Separate metadata fields that were available before the decision from retrospective settlement fields. Archive hashes and retrieval times do not by themselves prove historical availability. Where historic rule versions cannot be established, mark the limitation and avoid presenting those events as fully verified execution evidence.

8 Acquisition and coverage plan

Use staged, resumable acquisition. First run a small compatibility batch containing a few non-holdout historical days with confirmed station outcomes and model initialization times. Verify downloaded content, GRIB parsing, units, timestamps, and cache reuse. This batch tests plumbing, not profitability.

Next construct a coverage matrix by date and source. Record model cycles and horizons, observations, labels, contracts, price histories, execution evidence, and fees. Choose research dates from the intersection of usable sources; do not select dates because trades would have won. Retain all excluded dates with reasons.

Prefer at least two full years of overlapping data for development and seasonality analysis, with older NOAA observations available for climatology. This is a planning target, not an assertion that all archives overlap. If coverage is materially shorter, narrow the claims and obtain a data-adequacy decision before the economic campaign. Do not turn missing history into synthetic weather or prices.

Before bulk downloads, estimate bytes per date, total size, bandwidth and free storage from the pilot. Use small-region or indexed-range retrieval where the source supports it. Resume complete verified objects; incomplete downloads remain separate. Bound parallel requests, retry only transient failures with backoff, respect provider limits, and save a failed-object manifest.

The user has authorized downloading needed historical datasets. Use free public data first and proceed with ordinary reversible acquisition within available storage. Do not silently purchase data, create paid cloud infrastructure, or consume unbounded API budgets. Set explicit storage and transfer limits in configuration after measuring the sample. A cap is a stopping rule, not permission to delete unrelated files.

Once acquisition is complete, freeze a dataset manifest containing every included raw object, normalized file, hash, date range, source, and quality result. Assign a dataset version. Experiment workers receive that version only. Subsequent acquisitions create a new version and cannot silently change an active campaign.

9 Data contracts and prevention of look ahead

Define separate canonical tables for station metadata, observations, forecast values, target outcomes, contract rules, market observations, fee schedules, and dataset membership. Use UTC timestamps internally, while preserving the original time zone, climate date, and source strings needed to reconstruct conversions.

Required time fields are observation or valid time, forecast initialization where applicable, source issue time, first-known availability or a documented bound, retrieval time, revision identity, and simulated decision time. They answer different questions and must not be collapsed into a single date column.

For each simulated decision, include only data whose availability time is at or before that decision. If availability is uncertain, apply a predeclared conservative delay supported by evidence and rerun delay sensitivity tests. A GRIB forecast valid at noon may have been issued hours earlier; a corrected noon observation published later cannot be used at noon.

Register uniqueness keys. Examples are station plus report type plus valid time plus revision; model plus initialization plus lead plus variable plus grid cell; and market ticker plus event timestamp plus sequence or trade identifier. Duplicates must be reconciled rather than silently averaged. Preserve nulls and quality flags; do not convert missing temperature into zero.

Test units, geographic coordinates, finite values, chronological ordering, source coverage, and expected event-to-contract relationships. Validate probability mass and contract mappings separately. Outcome records reside in an evaluator-controlled location; discovery workers get only the development labels needed for their assignment.

Parquet tables and SQLite schemas should be versioned. Provide conversion functions for provider field changes, including decimal dollar strings and fractional quantities. Raw objects remain immutable. A deterministic dataset build must reproduce normalized outputs or explicitly explain unavoidable metadata differences.

10 Climate day and settlement mapping

The prediction target is the daily high named in the specific historical market contract. Verify the airport identity, reporting source, unit, climate-day boundaries, and resolution rule before mapping a model distribution to contracts. General city weather and airport settlement are not interchangeable [S3].

Kalshi's published daily-temperature guidance uses the NWS climate report and local standard time. For the LAX case, verify the historical contract's actual interval instead of blindly grouping by local civil midnight. Daylight-saving transitions need explicit tests. A standard-time day and a civil-calendar day can include different observations.

Model values can be instantaneous, averaged, or maxima over intervals. Inspect GRIB metadata rather than inferring semantics from filenames. Define how to handle forecast intervals that cross the target window; do not include an out-of-window maximum merely because its ending timestamp is inside the day. Keep discrete hourly maxima and interval maxima distinguishable.

Map a calibrated latent temperature distribution into the official reported outcome space using justified rounding and precision assumptions. Then integrate probability over each contract's exact bounds. Test strict and inclusive endpoints, open-ended tail contracts, YES and NO complements, and mutually exclusive bins. Do not mechanically normalize market asks into probabilities or assume independent contracts when all depend on the same daily temperature.

Observed high-so-far is useful but only a strict lower bound when the observation's station, precision, reporting window, and measurement definition are compatible with the settlement target. Otherwise carry measurement uncertainty instead of declaring incompatible outcomes impossible.

Reconcile NCEI daily summaries, NWS climate reports, and Kalshi expiration values. Investigate disagreements. For economic outcomes, use the actual settled market result and payout, with cancellations or exceptional resolutions handled explicitly. For forecast scoring, identify the agreed target label and retain any revisions separately.

11 Forecast models and calibration

Start with a seasonal climatology distribution built from prior station data, each available weather model separately, the collector's original fixed-width Gaussian, and a simple corrected ensemble. These baselines establish whether more complicated methods add information. Do not treat the collector's ensemble weights or 2 degree spread as validated.

For correction, fit station and lead-time bias and residual uncertainty using training folds only. Compare a small set of transparent models before nonlinear expansion. Candidate features include current temperature, high-so-far with measurement treatment, recent heating rate, wind, cloud state, forecast revisions, and coastal regimes. Define each feature at the simulated cutoff.

Use consistent decision times before and during the historical target day. Choose the candidate schedule before evaluating returns. Treat the schedule as a research parameter with a bounded search budget, not unlimited opportunities to select the most profitable minute after the fact.

Learn probability calibration within chronological training and validation. Check reliability overall, by forecast horizon, by probability range, and in the subset selected by the 10 percent rule. Avoid declaring calibration satisfactory from a handful of selected trades. GFS-derived products may share errors; estimate their incremental contribution rather than granting each an independent vote.

Report temperature MAE and bias, a proper distribution score such as CRPS or log score, contract-level Brier score, calibration diagnostics, sharpness, and performance against simple forecast and market-derived references. Define log-score clipping as a numerical convention before use and disclose it.

Agent reasoning proposes features and experiments. Fitted models and deterministic code produce authoritative predictions. Any LLM-extracted forecast-discussion feature must come from the historical text available at the cutoff, use a versioned extraction prompt, and preserve the extracted record for offline reruns. No model is promoted solely because an agent calls it meteorologically plausible.

12 Return definition and historical replay

Let q be the estimated probability that the purchased side pays one dollar, c the executable entry price per contract, f allocated entry fees, and s any additional applicable settlement costs. The initial strategy buys contracts and holds to settlement. Its estimated net return on entry outlay is:

```
expected_net_profit = q - c - f - s  
expected_net_return = (q - c - f - s) / (c + f)  
candidate_screen = expected_net_return >= 0.10
```

Use the YES probability for a YES purchase and its complement for NO. Apply actual size-dependent prices and fee rounding. Include spread and slippage in the modeled executable price and avoid counting them twice. A 10 percentage-point probability gap is not the same as 10 percent ROI.

For each hypothetical filled entry, compute historical simulated realized profit from the actual settlement payout less entry outlay and applicable costs. Report per-trade ROI and aggregate capital-weighted ROI as separate metrics. Aggregate ROI equals total simulated profit divided by total entry outlay; it is not an annual account return. A mean of individual trade ROIs answers a different question and must be labeled.

Implement an event-time replay with a frozen decision schedule, quote-age limit, order size, price convention, latency assumption, duplicate-entry rule, exposure cap and settlement policy. These are research settings, not permission to trade. Define conservative reference sizes and sensitivity cases before the campaign; do not quietly choose a size that improves reported returns.

Every decision record contains accepted or skipped status, reason, information cutoff, model and dataset versions, probability, costs, price source, size, expected ROI, and execution-evidence grade. The ledger links decisions to simulated entries and outcomes. Preserve zero-opportunity periods and losses. A strategy must be allowed to abstain when nothing meets the rule.

13 Execution evidence and honest gain reporting

Classify each replay scenario by the evidence supporting its hypothetical fill. **Grade A** uses sufficiently detailed historical quotes and sizes with a specified execution delay; it is an execution-aware simulation, still not proof of an actual fill. **Grade B** uses timestamped bid and ask summaries with limited depth or timing evidence, applies conservative assumptions, and reports an assumed-fill scenario. **Grade C** uses only prints or aggregated prices; it supports price-based sensitivity analysis but cannot substantiate an executable 10 percent edge.

Do not infer passive-order fills from a candle touching a limit. Queue priority, cancellations, available size, adverse selection, and event sequence matter. Exclude block trades from ordinary fill assumptions unless the strategy could have participated on equivalent terms. Never use the final settled market's displayed price as an earlier entry price.

If historical depth cannot be obtained, continue forecasting and limited price-based analysis, but mark execution adequacy as unresolved. A completed offline campaign may legitimately conclude that market execution evidence is insufficient. It may not label a weakly supported replay as proven realizable profit.

Run cost and timing sensitivity cases: historical fee schedule where available, present-cost alternative, wider spreads, delayed entry, stale quotes removed, and smaller available size. Freeze the cases before inspecting final results. Keep a count of how many qualifying opportunities disappear under each case.

Report net simulated PnL, entry outlay, trade count, independent weather-day count, wins and losses, average and capital-weighted returns, holding times, capacity, daily concentration, drawdown under an explicit capital model, and confidence intervals. Always show the selected date range and excluded-data ledger.

The term realized gains in this project means outcomes realized in a historical simulation from observed settlements. These are not gains earned in a brokerage account. The distinction must appear on every economic performance report. No numerical return result is currently established.

14 Evaluation and protection against overfitting

Select exact chronological partitions after the coverage audit and before discovery agents inspect evaluation outcomes. Prefer rolling-origin development folds, a separate chronological selection period, and a final protected test period. Keep all contracts and repeated forecasts from the same weather day together. Separate transformations, feature selection, calibration, and decision-rule fitting by fold.

A planning target is at least two years of usable overlapping inputs, with the latest protected interval long enough to cover meaningful variation. The actual final interval and minimum evidence requirements must be registered from data adequacy, not selected because its returns look favorable. If coverage is short, state the narrower inference and do not claim year-round robustness.

Use purging or temporal buffers when labels, forecast horizons or overlapping features can cross a partition boundary. Bootstrap or otherwise estimate uncertainty at the weather-day or suitable block level; thousands of related contracts are not thousands of independent weather events.

Log every hypothesis and parameter search, including failed attempts. Use the same score definitions across candidates. Compare the baseline, component changes, and combinations to measure incremental value. Evaluate the selected trading subset as well as the overall forecast distribution. Separate statistical evidence, practical significance, and claimed economic advantage.

The final evaluator receives one frozen candidate, code and data hashes, and locked settings. It calculates scores in a restricted process. Discovery agents do not receive final labels. Once final results influence development, the test is spent; future confirmation needs a new untouched period. Rerunning identical code to verify determinism is different from selecting another model against the same revealed outcomes.

Scientific outcomes are `SUPPORTED_WITH_LIMITATIONS`, `NO_IMPROVEMENT`, `INSUFFICIENT_DATA`, or `INVALIDATED`. Positive historical realized ROI alone does not prove true expected ROI is at least 10 percent. The final report must distinguish the estimated selection threshold from the strength of evidence about future expectation.

15 Research groups and the 36 tracks

Use six logical groups. A covers settlement and data; B covers forecast baselines; C covers intraday information; D covers contract probabilities and calibration; E covers market opportunities and execution; F covers independent challenge. Groups are assignments, not permanent processes. The controller schedules them within its worker limit.

The canonical track catalog is retained below. A track cannot run until its required data and evaluator are available.

R01 model ensembles	R02 KLAX model bias
R03 lead-time weighting	R04 forecast revisions
R05 marine-layer depth	R06 cloud clearing
R07 inversions	R08 onshore and offshore flow
R09 sea-breeze timing	R10 pressure gradients
R11 Santa Ana regimes	R12 seasonal coastal regimes
R13 current temperature	R14 high-so-far
R15 heating rate	R16 temperature curvature
R17 dew point	R18 wind observations
R19 cloud observations	R20 pressure observations
R21 measurement precision	R22 rounding uncertainty
R23 climate-day timing	R24 outcome revisions
R25 climatology	R26 historical analogs
R27 Bayesian combination	R28 regression
R29 gradient boosting	R30 probability calibration
R31 forecast-discussion text	R32 independent challengers
R33 model versus market	R34 bin consistency
R35 execution and timing	R36 adversarial and process review

For the first campaign, prioritize GFS/NBM baseline correction, mixture weights, spread calibration, and execution realism. Intraday updates, cloud-regime effects, forecast revisions and market response delays remain future hypotheses requiring additional historical inputs. Keep one alternative or falsification assignment in each follow-up phase.

Each researcher writes a claim, mechanism, required inputs, falsification condition, planned comparison, and expected artifact before implementation. A numerical success threshold must be registered by the evaluator, not negotiated after seeing results. Model-quality experiments and economic experiments have different scorecards.

Research group F applies to every track even though it also owns R36. It checks source timing, leakage, selection bias, missing prices, cost assumptions, same-day dependence, and replication. Several agents agreeing with one another is not an independent validation result.

16 Agent communication and deterministic control

The director recommends track assignments and follow-ups. Explorers propose hypotheses. Implementation workers write candidate code. Critics seek alternative explanations. Replicators rebuild the test from the specification. A synthesizer identifies useful combinations and contradictions. Software runs experiments, computes scores, and changes authoritative state.

Every task carries campaign ID, task ID, epoch, track, question, permitted inputs, prohibited inputs, dependencies, allowed outputs, runtime and token limits, retry count, and success or failure conditions. Workers return structured status, claims, evidence IDs, artifact paths, limitations, and suggested next steps.

Use independent initial hypotheses followed by local collaboration. Publish concise cross-group messages such as FINDING, QUESTION, CONTRADICTION, BLOCKER, or REQUEST_REPLICATION. Every factual finding links to an experiment artifact. Speculation remains labeled. The synthesizer reads structured summaries and can inspect underlying evidence when necessary; it does not consume every conversation by default.

Maintain separate registries for hypotheses, experiments, evidence, messages and decisions. Preserve parent-child relationships when ideas combine. Agents submit records through one transaction-safe writer. A hypothesis moves from proposed to tested, supported in development, replicated, challenged, and candidate only through recorded gates. Negative evidence is never overwritten by a positive summary.

The controller enforces allowed task types, concurrency, total resource budgets, timeout, bounded retries, deduplication, and legal state transitions. Model output is untrusted until its schema and requested operation are validated. Candidate code cannot modify evaluation logic or the controller's permissions. External text is data, never an instruction source.

Use isolated worker directories or worktrees for candidate code. Review changes before integration. Provide one durable queue with recovery after interruption; assign idempotency keys so a retry cannot count an experiment or spend budget twice. Reserve worst-case resources before dispatch and reconcile actual usage afterward. Log all exceptions and unresolved tasks.

17 Goal 1 implementation milestones

G1A Repository and dependency audit. Resolve collector access, preserve its pinned source, inspect code and dependencies, verify a supported environment, and inventory existing files. Deliver `COLLECTOR_AUDIT.md`, a dependency lock, source provenance, and working setup instructions. Do not claim a dependency works until imports and a tiny parse succeed.

G1B Target and historical acquisition. Verify settlement mapping, download and normalize necessary historical sources, extend beyond the bootstrap sample, and produce a coverage and exclusion report. Deliver a versioned manifest and documented storage estimate. Historical model forecasts and market prices are mandatory for the full economic research lane.

G1C Offline datasets and baselines. Implement as-of joins, climate-day handling, chronological partitions, baseline distributions, contract probability mapping, and deterministic scoring. Deliver saved baseline predictions and an audit showing they use no future information. Preserve the collector's original output as a comparison, not as accepted truth.

G1D Historical replay and reports. Implement decision records, cost and size assumptions, settlement ledger, execution grades, sensitivity cases, and a report showing simulated historical realized returns. The same fixture must produce known PnL by hand and in code. Missing execution evidence must appear in readiness and results.

G1E Controller and worker interfaces. Implement task schemas, registries, bounded scheduling, local messages, synthesis, critic and replication roles, isolated workspaces, and deterministic state. Demonstrate a complete small fixture campaign, including a failed hypothesis and a combined follow-up.

G1F Readiness verification. Run meaningful tests for time leakage, daylight saving, units, interval maxima, fees, probability mass, same-day grouping, denied holdout access, offline operation, budget exhaustion, worker crash, restart and duplicate results. Produce the readiness report with exact commands, environment, code hash, dataset version, evidence grade, passed checks and remaining blockers.

Goal 1 completes only when the runnable offline pipeline and controller pass their applicable gates. If the collector or required data remain inaccessible, report a specific blocker and continue independent work. Do not relabel a partially implemented system as ready. A fixture-only demonstration is not real-data readiness.

18 Goal 2 first offline multi agent campaign

Launch only against a Goal 1 readiness record whose code and dataset hashes match the current workspace. Verify that economic settings, research budget, exact date partitions, and execution-evidence limitations are explicit. No online acquisition occurs during the campaign; missing historical inputs return a blocked research task.

Use three research epochs, at most twelve hypotheses and no more than four agent slots including coordination. These are our engineering limits, not properties of OpenAI's system. The implemented local pilot below uses one inference process and zero paid API spending.

The implemented first pilot uses one local GPU inference process at a time, at most twenty-four model calls and a 393,216-token reservation budget. Each call has a 16,384-token context and a 768-token generation cap. One host-registered seed and up to nine model-proposed recipes are evaluated across the three rounds; duplicate recipes are not extra experiments. The closed recipe language permits only GFS/NBM mixture weights, three mean-calibration methods, three spread methods and enumerated spread adjustments. The broader track catalog is a roadmap rather than a claim that all thirty-six tracks run in this pilot. Paid API spending is zero.

Before any historical performance is inspected, the development screen is fixed: at least two percent CRPS improvement over the best registered baseline, Brier deterioration no greater than 0.005, at least thirty simulated entries, at least ten percent capital-weighted simulated return, nonnegative return under the specified higher-cost stress, independent verification of primary opportunity selection and an explicit nonrejecting critic. Rank candidates by CRPS, Brier and identifier. A qualifying champion is frozen for one final evaluation. No qualifying candidate means a valid no-improvement report with the protected interval unused. See configs/pilot.json and docs/CAMPAIGN_PROTOCOL.md for exact definitions and limits.

Before dispatch, record all limits in the manifest: six hours of campaign wall time, at most two retries for transient failures, and the task, experiment and local-compute budgets. The controller may stop early but may not exceed these limits.

Epoch 1 independently proposes recipes within the permitted mixture, mean-calibration and spread-calibration families while a separate audit checks data and replay assumptions. Later rounds use recorded evidence to propose parent-linked combinations and a simpler or conflicting challenger. Every tested recipe receives independent numerical verification; the final synthesis and critic determine whether any candidate meets the frozen development gates. Role assignments run in waves with one local inference process at a time. Intraday forecast-update research is outside this first dataset's resolution.

Record a STARTED event only after a real worker accepts a task. Check that at least one real historical experiment, one evidence exchange and synthesis decision, and the required replication or rejection path complete. A candidate advances to final evaluation only if its predefined development gates pass. Otherwise finish with no improvement or insufficient evidence and explain why.

Goal 2 ends with a complete campaign ledger, baseline and candidate results, an offline historical PnL report, resource usage, an independent review, and a clear next decision. Technical failure before the cycle completes is not success. Keep the system offline after completion, regardless of reported returns.

19 Runtime operations and proposed command interface

The following command names are an interface specification to implement, not executable commands supplied by this document. The implementation may choose a different package name but must document equivalent operations and demonstrate them before declaring readiness.

```
klax-lab doctor
klax-lab acquire --plan configs/historical_sources.yaml
klax-lab validate-data --manifest <dataset-version>
klax-lab build-dataset --policy configs/evaluation.yaml
klax-lab baseline --offline --dataset <version>
klax-lab replay --offline --policy configs/replay.yaml
klax-lab readiness --offline
klax-lab campaign start --offline --config configs/pilot.yaml
klax-lab campaign status <campaign-id>
klax-lab campaign stop <campaign-id>
klax-lab campaign resume <campaign-id>
klax-lab report <campaign-id>
```

The acquisition command has a historical date plan and resumable manifest. All experiment commands require local inputs. A campaign can stop cleanly, preserve completed work, and resume without duplicate jobs. No command should implement order placement or start an online service.

The implemented equivalents are documented in docs/OFFLINE_COMMANDS.md. Campaign pause/resume preserves identity, evidence and spent budgets; an interrupted process requires explicit review before resuming. Unknown worker usage is charged at its reserved limit, and pauses do not reset the original wall-time budget. A completed campaign can be reused for reporting without dispatching new researchers or replacing its readiness record.

Keep source code under src/klax_lab/, tests under tests/, domain configuration under configs/, schemas under schemas/, and instructions under docs/. Use ignored data/raw/, data/normalized/, data/frozen/, and runs/ directories for bulk data and artifacts. Store small dataset manifests and configuration versions in Git. Protect final labels with actual access controls appropriate to the backend.

Record interpreter and native-library versions, source revisions, seeds and hashes. Provide a reproducible environment and a cache restoration procedure. Back up registries and manifests before migrations. Never delete source evidence to satisfy a quota without an explicit retention policy and a verified replacement.

An economic report must be reproducible from saved predictions and ledger rows without invoking an LLM or downloading data. A worker can fail while the report remains auditable. Logs distinguish unavailable data, invalid experiment, coding error, insufficient budget and scientific rejection.

20 Acceptance evidence and final handoff

The Goal 1 readiness report must include an implementation inventory, collector audit, historical source coverage, missing-data and exclusion ledger, verified station and contract mapping, as-of timing checks, split manifest, baseline metrics, replay fixtures, network-denial check, isolation and holdout tests, resource limits, and exact runnable commands. Every pass links to an artifact or test output. No pass can rest on agent prose alone.

The Goal 2 report must show the actual campaign ID and start/end times; all proposed, executed and rejected hypotheses; communication and synthesis records; candidate code and dataset hashes; independent replication; final-test usage; economic settings; expected versus simulated realized returns; uncertainty; capacity and frequency; and execution-evidence grade. Include the strongest contrary evidence, not just the winning model.

Required comparison tables are baseline versus candidate forecast scores, selected-opportunity calibration, cost sensitivity, and per-day simulated PnL. Show capital-weighted return and unweighted mean trade return separately. Distinguish independent weather days from contract count. A plot of cumulative simulated profit needs its sizing and capital assumptions next to it. The first pilot has hourly price summaries and a single 00 UTC weather run: it cannot validate subhour execution latency or intraday forecast-update advantages. The registered sixty-second entry delay is an assumption, not a measured latency result. Preserve this explicit data-resolution limitation rather than inventing a latency sweep from unavailable prices; those research tracks require a later historical dataset and separate validation plan.

There are three separate achievements: the system works; a method improves forecast or replay performance on unseen historical data; and evidence supports a persistent future economic edge. Goal completion establishes the first and evaluates the second. It cannot guarantee the third. A positive backtest is a reason for review, not automatic online deployment.

The first pilot's configuration is now recorded in `configs/evaluation.json` and `configs/pilot.json`: chronological date partitions, evidence criteria, quantity, decision time, stale-quote limit, fee proxy, download caps and local worker backend. The preferred collector source has been recovered and audited. Exact historical fee applicability and executable depth remain unresolved data limitations, not silently assumed verified facts. Routine choices are documented; further credentials, purchases or platform changes are outside this first pilot unless separately requested.

The user has authorized execution of Goal 1 followed by Goal 2. The historical source audit and local implementation are in progress, with the weather archive still downloading at this revision. The finite batch may advance only when its recorded prerequisites pass. Current execution evidence and any remaining blockers belong in `docs/IMPLEMENTATION_STATUS.md` and the machine-readable run state; a plan, queued job or synthetic rehearsal is not a completed research campaign.

21 Sources and provenance

[S1] Apply Navier Stokes Method. User's prior planning conversation.

<https://chatgpt.com/c/6aa9e345-d790-83e8-a399-b32ae86b65bf>

[S2] ciel-ravencourt weather_data_collector. Private repository README inspected in the user's authenticated browser; source revision a82f8aee0afecf568b3ce16f339c0bd7201773b8.

https://github.com/ciel-ravencourt/weather_data_collector

[S3] Kalshi Weather Markets. Settlement source and climate-day guidance; specific historical contract rules govern. <https://help.kalshi.com/en/articles/13823837-weather-markets>

[S4] Kalshi Historical Data and historical market/trade endpoints. Coverage must be verified by actual retrieval.

https://docs.kalshi.com/getting_started/historical_data ;

<https://docs.kalshi.com/api-reference/historical/get-historical-markets> ;

<https://docs.kalshi.com/api-reference/historical/get-historical-trades>

[S5] Kalshi Fees and Orderbook. Fee schedule and executable-price context.

<https://help.kalshi.com/en/articles/13823805-fees> ; <https://help.kalshi.com/en/articles/13823828-the-orderbook>

[S6] NOAA NBM archive listing. Candidate archived model source. <https://registry.opendata.aws/noaa-nbm/>

[S7] NOAA HRRR archive listing. Candidate additional short-range forecast source.

<https://registry.opendata.aws/noaa-hrrr-pds/>

[S8] NOAA NCEI station details for Los Angeles International Airport, USW00023174. Station identity and daily archive. <https://www.ncei.noaa.gov/cdo-web/datasets/GHCND/stations/GHCND:USW00023174/detail>

[S9] NOAA NWS API documentation. Source interfaces and time-aware product discovery; historical availability is not guaranteed by a current API. <https://www.weather.gov/documentation/services-web-api>

[S10] OpenAI On the Navier Stokes Millennium Prize Problem. Organizational inspiration, not a specification of our controller. <https://openai.com/index/navier-stokes-solution/>

Local primary evidence: docs/bootstrap-download-manifest.json and the three raw bootstrap files described in Section 3. Their source hashes distinguish observed data from the proposed architecture. This document contains no measured strategy-return result.

Review versions of this document may be shared with the user, but the private collector source and credentials must not be published. Keep the master Markdown and rendered PDF synchronized when changing a goal, scope boundary, acceptance rule, or source status.